Fiscal Year:	FY 2008	**Task Last Updated:**	FY 11/30/2009
PI Name:	Christian, James Ph.D.		
Project Title:	Tissue-Equivalent Radiation Dosimeter-On-A-Chip		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **ARS**:Risk of Acute Radiation Syndromes Due to Solar Particle Events (SPEs)		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	JChristian@rmdinc.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	(616) 668-6897
Organization Name:	Radiation Monitoring Devices, Inc.		
PI Address 1:	44 Hunt Street		
PI Address 2:			
PI Web Page:			
City:	Watertown	**State:**	MA
Zip Code:	02472-4699	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	SBIR Phase II
Start Date:	02/11/2008	**End Date:**	02/10/2010
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Cucinott1a, Francis	**Contact Phone:**	281-483-0968
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ08JA55C		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	Available digital dosimeters are bulky and unable to provide real-time monitoring of dose for space radiation. The complexity of space-flight design requires reliable, fault-tolerant equipment capable of providing real-time dose readings during a mission, which is not feasible with the existing thermo-luminescent dosimeter (TLD) technology, especially during extravehicular activities (EVA). Real-time monitoring is important for low-Earth orbiting spacecraft and interplanetary space flight to alert the crew when Solar Particle Events (SPE) increase the particle flux of the spacecraft environment. The Phase-II project will design and fabricate a prototype Dosimeter-on-a-Chip (DoseChip) for personal dosimetry comprised of a tissue-equivalent scintillation crystal coupled to a solid-state photomultiplier (SSPM). The ubiquitous nature of CMOS technology provides a standardized development platform, and the ability to integrate the supporting electronics into a miniature, lightweight design. The DoseChip provides a tissue-equivalent response to the relevant energies and types of radiation for low-Earth orbit and interplanetary space flight to the moon or Mars and will be sensitive to the dose rates and particle fluxes of ambient Galactic Cosmic Rays (GCR) to the higher rates of		

	major SPE. The DoseChip will complement the existing Crew Passive Dosimeters by providing real-time dosimetry and as an alarming monitor for SPE.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Inexpensive radiation dosimeters are in general demand for commercial applications, such as personnel or waste monitoring, other applications include border monitoring for homeland security and protecting satellites and ground-based equipment from solar flares. The ability to inexpensively mass-produce these devices creates an entirely new market for arrays of distributed sensors. As high altitude commercial flight become more prevalent, active dosimeters can provide the redundant safeguards and information required to protect companies from unnecessary litigation and passengers from hazardous radiation conditions.
Task Progress:	New project for FY2008. Reporting not required for this SBIR Phase 2 project.
Bibliography Type:	Description: (Last Updated:)